

Neural Network Equity Valuation

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A study of how financial fundamentals, digital innovation and ownership structure can inform the valuation of Chinese listed software companies. This public edition summarizes the original undergraduate thesis and reproduces selected reported results. It is an academic valuation study, not a trading strategy or a live investment track record.

Research question

Can a feedforward neural network capture nonlinear relationships between company characteristics and observed equity market value? The study focuses on software businesses, where conventional financial measures may not fully describe innovation and ownership-related differences.

Study dimension	Documented scope
Population and period	388 A-share software companies; 2011–2023
Input variables	14 financial and non-financial indicators
Factor analysis	Four factors; cumulative explained variance of 71.854%
Reported test performance	R^2 0.9800; MSE 0.0099 on the study's transformed scale
Company case	Yonyou Network: estimated RMB 26.84bn versus RMB 25.52bn market value

Work demonstrated

The project connects financial-statement interpretation with data preparation, multivariate analysis and a neural-network valuation exercise. It documents sample selection, outlier treatment, feature scaling, factor diagnostics and the translation of model output into a company valuation comparison.

How to read the evidence

The figures above are transcribed from the original thesis. They have not been independently re-estimated for this release. The original training code, fitted weights and exact train/test split are not included because a complete implementation was not recovered from the reviewed materials. The repository distinguishes documented results from reproducible computation.

Method and validation boundaries

Inputs and preparation

The thesis describes financial and non-financial data sourced from CSMAR, exclusion of ST/*ST observations and incomplete samples, two-sided 1% winsorization of continuous variables, and Z-score scaling. The 14 indicators cover profitability, receivables turnover, liquidity, growth, digital patents and shareholder concentration. Licensed raw observations and company-level extracts are intentionally absent from this release.

Model specification in the detailed methods

The detailed specification describes 14 input neurons, one hidden layer containing four neurons, tanh activation and one output representing equity market value. Exploratory factor analysis supports the choice of four hidden units. Introductory wording elsewhere in the thesis differs on network depth; the detailed specification is reported here without asserting that the trained architecture has been recovered.

Split	MSE	RMSE	MAE	R ²
Training	0.0098	0.0993	0.0801	0.9809
Test	0.0099	0.0990	0.0777	0.9800

Source: original thesis, Table 3-6. Values are reproduced as reported. Rounding or transcription may explain small inconsistencies between MSE and squared RMSE. These errors should not be interpreted as currency-unit prediction errors.

Case interpretation

The thesis reports a RMB 1.32bn difference between the estimated and observed market values of Yonyou Network. Dividing 1.32 by 25.52 gives approximately 5.17%. This is a single academic case comparison. It does not establish prospective accuracy, mispricing, or an investable opportunity.

Further checks needed before investment use

A complete replication would need the original training implementation and weights, date-stamped source data, disclosure-lag controls, and the exact sample/split definition. Preprocessing must be fitted only on training data. Chronological and company-disjoint tests, baseline comparisons and sensitivity checks would be needed before making claims about generalization or investment usefulness.

Selected original evidence

The following short Chinese excerpts and numerical results come from the original thesis. The public edition omits the student number, class details, supervisor information, acknowledgements, document metadata and database exports.

Original research title

基于前馈神经网络的数字企业价值评估与应用——以软件开发行业为例

Sample selection excerpt

“本文将以2011-2023年中国A股软件开发行业上市公司为初始观测样本，基于证监会行业分类标准（2023版）选定行业代码为I65的上市公司，经匹配和清洗后获得有效样本388家。”

Factor analysis excerpt

“在SPSS因子分析中共提取出4个有效的因子，且所有被提取因子特征根值均大于1。在经过旋转后，所提取的4因子的累积方差解释率达到71.854%。”

Original metric transcription

Item	Reported value
Test R ²	0.9800
Test MSE	0.0099
Yonyou observed equity value	RMB 25.52 billion
Yonyou estimated equity value	RMB 26.84 billion
Relative valuation difference	5.17%

Evidence scope

This edition makes the research question, chosen inputs and reported outputs inspectable while preserving data rights and personal privacy. It is not a certificate of independent replication. See docs/EVIDENCE_AND_LIMITATIONS.md for the source mapping and unresolved replication requirements.